STUDENT RECORD SYSTEM

SYSTEM VALIDATION REPORT

Application acceptance, integrated workflows, and performance validation

Document	TEST-SYS-02
Project	Student Record System
Course	Software Engineering Mini Project
Team	Team 11
Implementation	C++17 application with SQLite persistence
Report basis	Recorded project test results and coverage.xml evidence available in the project conversation
Report date	10 November 2026

Prepared as a final testing report for the project documentation package.

1. Purpose

This report documents validation of complete Student Record System workflows through automated application acceptance scripts, together with the recorded performance benchmark results. System-level checks exercise the compiled application and verify visible outcomes and role restrictions.

2. Validation Environment

Component	Recorded configuration
Application	Student Record System — C++17
Persistence	SQLite 3
Acceptance scripts	verify_attendance.sh; verify_reports.sh
Automation	GitHub Actions
CI platforms recorded	Ubuntu, macOS and Windows
Performance tool	performance_test.cpp

3. Attendance Workflow Validation

The attendance acceptance script verify_attendance.sh is integrated into the CI workflow and executes against the compiled application.

Test ID	Validation scenario	Expected verification	Recorded status
TC-ATT-01	Record attendance	Attendance operation succeeds for the selected student and subject	PASS in successful CI run
TC-ATT-02	Calculate attendance percentage	Displayed percentage matches the test records	PASS in successful CI run
TC-ATT-03	Update attendance	Updated attendance is reflected in the calculated percentage	PASS in successful CI run
TC-ATT-04	Student access restrictions	Unauthorized student modification is denied	PASS in successful CI run

4. Academic Reports Workflow Validation

The academic reporting acceptance script verify_reports.sh is integrated into the CI workflow and exercises the compiled application.

Test ID	Validation scenario	Expected verification	Recorded status
TC-RPT-01	Generate full academic report	Expected report sections are displayed	PASS in successful CI run
TC-RPT-02	Generate grade sheet	Grade-sheet output is displayed	PASS in successful CI run
TC-RPT-03	Generate attendance summary	Attendance summary is displayed	PASS in successful CI run
TC-RPT-04	Role-based report access	Report permissions are enforced	PASS in successful CI run

5. Performance Validation

Performance testing used a synthetic dataset and recorded five warm-up runs followed by 100 measured runs. The acceptance target recorded for each benchmark was P95 below 3,000 ms.

Dataset parameter	Value
Students	5,000
Subjects	10
Attendance records	250,000
Warm-up runs	5
Measured runs	100

Operation	Average (ms)	Median (ms)	P95 (ms)	Result
Student listing	0.374689	0.3614	0.4605	PASS
Student search	0.002018	0.0020	0.0021	PASS
Attendance summary	0.01003	0.0100	0.0102	PASS

Interpretation: all three recorded P95 values are below the 3,000 ms target. These figures are SQLite query execution times, not end-to-end application response times.

6. CI Execution Evidence

• The attendance and academic-report acceptance steps are included in the GitHub Actions workflow.

• The recorded project results state that the updated workflow completed successfully after Pull Request #31 was merged into main.

• The Code Coverage (Ubuntu) job also completed successfully and uploaded its coverage artifact.

• This report does not invent a run ID, execution timestamp, or assertion count; those details should be taken from the corresponding GitHub Actions run page.

7. Overall Validation Result

The recorded successful CI runs support a PASS result for the attendance and academic-report acceptance workflows. The recorded performance benchmark results also meet the stated P95 target. Validation is limited to the scenarios and dataset described above.

8. Limitations

• The benchmark is a database-query microbenchmark and does not establish full UI or end-to-end response latency.

• CI pass status is based on the recorded successful workflow evidence; this report does not include a specific GitHub run identifier.

• This report does not claim validation of every SRS requirement.

9. Conclusion

The system-level acceptance scripts cover key attendance and academic-report workflows, including access restrictions. The recorded benchmark results meet the specified P95 threshold for the three measured database operations.